

SkyTouch: Augmenting Situational Awareness for Multi-sUAS Pilots through Multi-Modal Haptics

Johanna Olesk

Department of Computer Science and Engineering
University of Notre Dame
Notre Dame, Indiana, USA
jolesk@nd.edu

Arturo Miguel Russell Bernal

Department of Computer Science and Engineering
University of Notre Dame
Notre Dame, Indiana, USA
arussel8@nd.edu

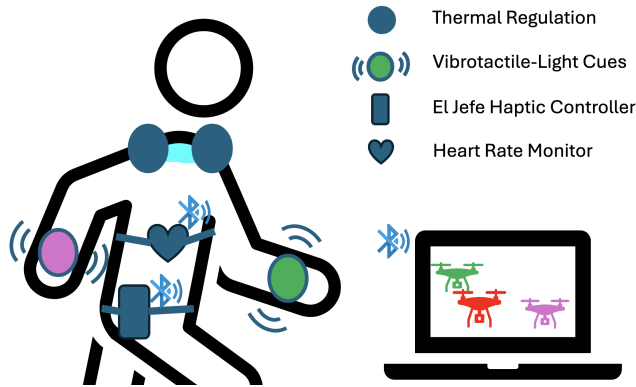


Figure 1: SkyTouch haptic feedback system supports sUAS operators by providing thermal feedback via Dots placed on the back of the neck for stress regulation, and wrist-worn Dots that deliver vibrotactile-light cues to enhance situational awareness and direct attention to critical alerts.

Abstract

Small Unmanned Aircraft Systems' (sUAS; e.g., drones) pilots are often overwhelmed by data-heavy interfaces in critical missions such as Search and Rescue (SAR), resulting in high cognitive load and increased risk of error, especially in multi-sUAS scenarios. We introduce *SkyTouch*, a wearable haptic system that enhances the human-sUAS interface by translating complex data streams into an intuitive, peripheral language of sensation, establishing a novel "felt" connection between pilot and sUAS. Our system delivers thermal feedback on the neck to help regulate operator's emotional state and wrist-based vibrotactile-light cues to guide visual attention toward critical alerts without cognitive overhead. Through this multi-modal dialogue of feeling, *SkyTouch* moves beyond basic alerts to support a deeper situational awareness and foster more resilient human-machine collaboration.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI); Haptic devices; User interface design.**

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Keywords

Haptics, sUAS, Drone Pilots, Human-Machine Interaction, Cognitive Load, Wearable Computing, Multimodal Feedback

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1 Introduction

High-stakes small Unmanned Aircraft System (sUAS; e.g., drone) operations, such as search-and-rescue (SAR), rely heavily on the human operator's ability to make critical decisions under pressure and maintain situational awareness over long periods of time [2, 6, 12]. A core challenge in these missions is the overwhelming volume of visual and auditory input that operators must process [3, 15, 21]. They are required to maintain continuous awareness of obstacles, battery levels, and navigational data, often leading to cognitive overload and elevated stress, which increases the risk of errors or missed cues [8, 10, 22]. While operators typically control a single drone, many real-world missions require the coordination of multiple sUAS simultaneously, further intensifying cognitive demands and exacerbating interface constraints [5].

To address these limitations in multi-sUAS missions, we present *SkyTouch* (see Fig. 1), a haptic interface that extends beyond existing approaches that simply translate telemetry to vibrations [13, 17]. Our system features two key contributions: **(1) orthogonal multi-modal mapping** that assigns each feedback type (thermal, vibration, light) to non-competing cognitive channels, preventing sensory interference; **(2) thermal regulation** that combines sUAS states with operator physiology (e.g., heart rate) to deliver interventions only when necessary, using evidence-based cooling protocols to enhance sustained attention before errors occur [19].

For this proposal, we situate *SkyTouch* in an established DroneResponse (DR) ecosystem [4, 5, 20] designed for the deployment of multiple semi-autonomous sUAS. DR comprises three principal components: multi-sUAS equipped with an onboard autonomous pilot (mCube), a Ground Control Station (GCS), and a Human Operator (HO; see Fig. 2). mCubes interface with sUAS flight controllers through serial wire connections and communicate with the GCS via MQTT and UDP protocols over a Doodle Labs mesh radio network. The HO supervises missions through a custom-GUI in the GCS and can override autonomy using a Remote Controller (RC) if needed.

Although FAA Part 107 [9] requires one pilot per sUAS, DR operates under a waiver allowing one HO to oversee multiple sUAS,

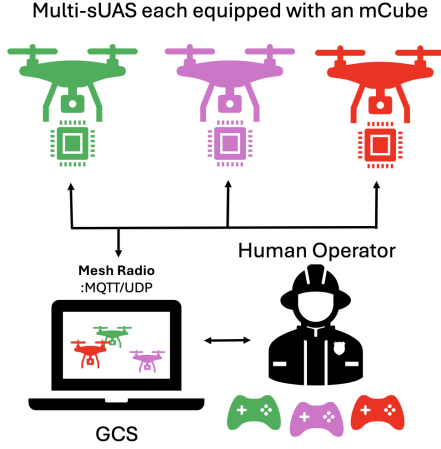


Figure 2: DroneResponse ecosystem, emphasizing the color-coded requirement for multi-sUAS deployment.

provided that each sUAS, its mCube, RC, and GUI representation are uniquely and clearly color-coded. Although this setup permits a single operator, in practice, multiple team members often oversee missions to distribute cognitive load. Based on prior live test missions, HOs frequently experience high stress and interpersonal conflict, often due to a loss of situational awareness [1]. To address these challenges, we propose *SkyTouch* by integrating a haptic feedback system that enhances situational awareness and helps regulate the HO’s stress levels during missions.

2 SkyTouch System Design

SkyTouch integrates the DataFeel Dev Kit [7], consisting of four Haptic Dots and the El Jefe™ wearable controller, into the DR ecosystem. *SkyTouch* is designed around the principle of orthogonal multimodal feedback where each Dot modality (thermal, vibration, and light) serves a non-overlapping role to reduce sensory interference and cognitive load. We position two Dots at the C7 vertebra (neck base), an area with high thermal sensitivity [14], to deliver thermal feedback for stress regulation. The other two Dots are positioned on the HO’s wrists to provide vibrotactile and visual alerts, enhancing situational awareness during multi-sUAS missions.

Cognitive Cooling (Slow Channel). The thermal feedback modality is designed to support the HO’s emotional state through the DataFeel Dots positioned at the back of the neck. An intervention is triggered when telemetry from a high-risk sUAS coincides with elevated heart rate data, indicating cognitive stress. In response, the system activates the Cognitive Cooling protocol, gradually reducing the Dot temperature to a calming but non-distracting 20°C [11]. Based on established research [16, 19], this subtle thermal intervention is intended to improve performance on tasks requiring sustained attention. Unlike an alert, it functions as a subconscious background regulator: a gentle, reassuring touch from the system when the cognitive pressure increases.

Vibrotactile Cues (Fast Channel). Vibration from two wrist-mounted Dots is reserved for high-priority events that require

immediate attention. We leverage the broad frequency range of the DataFeel Dots to create distinct haptic textures.

Tactile Alert. To guide visual attention, the wrist Dots serve as directional pointers to the operator’s visual interface. When a critical event occurs, such as low battery, the left wrist vibrates if the affected sUAS appears on the left side of the GCS’s GUI; the right wrist vibrates for right-side alerts. This spatial mapping quickly directs the operator’s gaze to the relevant area of the screen, allowing fast identification of the sUAS requiring attention.

Haptic Confirmation. To reduce uncertainty and reinforce command acknowledgment, a low frequency, crisp “thump” sensation provides confirmatory feedback, e.g., when a drone confirms execution of a critical command such as “Return to Launch” (RTL). This feature provides the HO a feedback loop without requiring visual attention, helping to reduce cognitive load.

Color-ID. Visual LED cues are coupled with high-priority vibrotactile feedback to help the HO quickly identify which sUAS triggered the alert. Since each sUAS in the DR ecosystem is assigned a unique color, these light-based cues make it easy to visually associate an alert with the corresponding sUAS. When a vibrotactile signal is active on the wrist Dot, a flashing LED displays the color of the sUAS requiring attention. If multiple sUAS raise alerts simultaneously, the LED cycles through their corresponding colors. This allows the HO to immediately direct focus to the appropriate sUAS without spending time identifying the source of the alert.

3 Implementation and Feasibility

SkyTouch is implemented as a microservice deployed in the GCS, integrating seamlessly into DR’s MQTT communication network. It uses Bluetooth as the interface between GCS, El Jefe, and the complementary heart rate sensor. While field tests of the full system are planned, the live in-door demo relies on DR’s simulation interface, with the sUAS simulated in Gazebo [18], an established high-fidelity physics simulator, ensuring real-time, reliable performance for the UIST demonstration. Leveraging DR’s existing ecosystem, the DataFeel SDK, and standard Bluetooth protocols makes this project highly feasible within the contest timeframe.

4 Conclusion

We have proposed an orthogonal, multi-modal feedback system for multi-sUAS operators, providing thermal, vibrotactile, and visual cues to help regulate their stress, reduce cognitive load and increase situational awareness. This abstract demonstrates the potential of haptic feedback to support HO in high-stakes missions (e.g., SAR). Beyond this use case, *SkyTouch* envisions broader integration of haptics across diverse SAR and sUAS stakeholders, not only allowing a Touch from the Sky, but also envisioning a Touch to the Sky by allowing direct manipulation of sUAS through haptics.

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References

- [1] Ankit Agrawal, Sophia J Abraham, Benjamin Burger, Chichi Christine, Luke Fraser, John M Hoeksema, Sarah Hwang, Elizabeth Travník, Shreya Kumar, Walter Scheirer, et al. 2020. The next generation of human-drone partnerships: Co-designing an emergency response system. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*. 1–13.
- [2] Arturo Miguel Russell Bernal and Jane Cleland-Huang. 2023. Hierarchically organized computer vision in support of multi-faceted search for missing persons. In *2023 IEEE 17th International Conference on Automatic Face and Gesture Recognition (FG)*. IEEE, 1–7.
- [3] Arturo Miguel Russell Bernal, Jane Cleland-Huang, and Walter Scheirer. 2025. Psych-Occlusion: Using Visual Psychophysics for Aerial Detection of Occluded Persons During Search and Rescue. In *2025 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*. IEEE, 3383–3395.
- [4] Jane Cleland-Huang. 2023. DroneResponse Quarry Demonstration. <https://www.youtube.com/watch?v=DyKqxkesgg0> Last Accessed July 18, 2025.
- [5] Jane Cleland-Huang, Theodore Chambers, Sebastian Zudaire, Muhammed Tawfiq Chowdhury, Ankit Agrawal, and Michael Vierhauser. 2024. Human-machine teaming with small unmanned aerial systems in a mape-k environment. *ACM Transactions on Autonomous and Adaptive Systems* 19, 1 (2024), 1–35.
- [6] Jane Cleland-Huang, Pedro Antonio Alarcon Granadeno, Arturo Miguel Russell Bernal, Demetrius Hernandez, Michael Murphy, Maureen Petterson, and Walter Scheirer. 2025. Cognitive Guardrails for Open-World Decision Making in Autonomous Drone Swarms. *arXiv preprint arXiv:2505.23576* (2025).
- [7] DataFeel. 2025. DataFeel Dev Kit. <https://www.datafeel.com>. Last Accessed August 05, 2025.
- [8] Alexandre Duval, Anita Paas, Abdalwhab Abdalwhab, and David St-Onge. 2023. The eyes and hearts of UAV pilots: observations of physiological responses in real-life scenarios. In *2023 32nd IEEE International Conference on Robot and Human Interactive Communication (RO-MAN)*. IEEE, 2352–2358.
- [9] Department of Transportation Federal Aviation Administration (FAA). 2025. PART 107—SMALL UNMANNED AIRCRAFT SYSTEMS. Published in the Electronic Code of Federal Regulations (eCFR). <https://www.ecfr.gov/current/title-14/chapter-1/subchapter-F/part-107> Last Accessed July 18, 2025.
- [10] Pedro Antonio Alarcon Granadeno, Arturo Miguel Russell Bernal, Md Nafee Al Islam, and Jane Cleland-Huang. 2024. An environmentally complex requirement for safe separation distance between uavs. In *2024 IEEE 32nd International Requirements Engineering Conference Workshops (REW)*. IEEE, 166–175.
- [11] Barry G Green. 2009. Temperature perception on the hand during static versus dynamic contact with a surface. *Attention, Perception, & Psychophysics* 71, 5 (2009), 1185–1196.
- [12] Sophie Hart, Victoria Steane, Seth Bullock, and Jan M Noyes. 2022. Understanding human decision-making when controlling UAVs in a search and rescue application. In *Human Interaction & Emerging Technologies (IHET 2022): Artificial Intelligence & Future Applications*. AHFE International.
- [13] Felix Huppert, Gerold Hoelzl, and Matthias Kranz. 2021. GuideCopter-A precise drone-based haptic guidance interface for blind or visually impaired people. In *Proceedings of the 2021 CHI conference on human factors in computing systems*. 1–14.
- [14] Manuela Jungmann, Shervin Vencatachellum, Dimitri Van Ryckeghem, and Claus Vögele. 2018. Effects of cold stimulation on cardiac-vagal activation in healthy participants: randomized controlled trial. *JMIR formative research* 2, 2 (2018), e10257.
- [15] Sara Ljungblad, Yemao Man, Mehmet Aydın Baytaş, Mafalda Gamboa, Mohammad Obaid, and Morten Fjeld. 2021. What matters in professional drone pilots’ practice? An interview study to understand the complexity of their work and inform human-drone interaction research. In *Proceedings of the 2021 CHI conference on human factors in computing systems*. 1–16.
- [16] Maohui Luo, Zhe Wang, Hui Zhang, Edward Arens, Davide Filingeri, Ling Jin, Ali Ghahramani, Wenhua Chen, Yingdong He, and Binghui Si. 2020. High-density thermal sensitivity maps of the human body. *Building and environment* 167 (2020), 106435.
- [17] Luis Moreno, Miguel Altamirano Cabrera, Muhammad Haris Khan, Issatay Tokmurziyev, Yara Mahmoud, Valerii Serpiva, and Dzmitry Tsetserukou. 2025. Fly-Haptics: Flying Multi-contact Haptic Interface. *arXiv preprint arXiv:2505.02582* (2025).
- [18] Open Robotics. 2025. Gazebo. <https://gazebo.org/home>. Last Accessed August 05, 2025.
- [19] Vincent Philippe, Jeanne Hecquard, Emilie Hummel, Ferran Argelaguet, Marc Macé, Valérie Gouranton, Claudio Pacchierotti, Anatole Lécuyer, and Justine Saint-Aubert. 2024. Cool me down: Effects of thermal feedback on cognitive stress in virtual reality. In *International Conference on Human Haptic Sensing and Touch Enabled Computer Applications*. Springer, 433–439.
- [20] Drone Response. 2025. Drone Response: To Protect and Preserve Life. <https://droneresponse.ai> Last Accessed July 18, 2025.
- [21] Arturo Miguel Russell Bernal, Walter Scheirer, and Jane Cleland-Huang. 2024. NOMAD: A natural, occluded, multi-scale aerial dataset, for emergency response scenarios. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*. 8584–8595.
- [22] Rajiv Kumar Saini, MSVK Raju, and Amit Chail. 2021. Cry in the sky: Psychological impact on drone operators. *Industrial psychiatry journal* 30, Suppl 1 (2021), S15–S19.